

Probability Theory

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An overview of probability using measures. We will denote probability measures defined over σ -algebras with $\mathbb{P}$ and probability functions defined over some sample space Ω or $\mathbb{R}$ with P or p . When we say countable, we mean finite or countably infinite. I have used resources from:

1. Rick Durret's *Elementary Probability and Probability* textbooks.
2. Dr. Krishna's *Probability Foundation for Electrical Engineers* lectures at IIT.
3. Various quant interview books and websites for examples.

1 Jan 12

Example 1.1 (Branching Processes)

Let Z_n be the number of people in the n th generation, and $X_i^{(n+1)}$ be the number of offspring the i th person in the n th generation has. Assume $X_i^{(n+1)}$ is iid with distribution X . Then, we have

$$Z_{n+1} = X_1^{(n+1)} + X_2^{(n+1)} + \dots + X_{Z_n}^{(n+1)}, \quad Z_1 = 1 \quad (1)$$

What is the extinction probability π for given X ?

$$\pi = \mathbb{P}(\exists n \text{ s.t. } Z_n = 0) \quad (2)$$

Some ideas is that we can try to compute $\mathbb{E}[X]$, use a Markov chain. We can also use a generating function, which is defined

$$f(\theta) = \mathbb{E}[\theta^X] = \sum_{k=0}^{\infty} \theta^k \mathbb{P}(X = k) \quad (3)$$

Then, we can define $f_n(\theta) = \mathbb{E}[\theta^{Z_n}]$. Let $\pi_n = \mathbb{P}(Z_n = 0)$. By the dominated convergence theorem,

$$\pi = \lim_{n \rightarrow \infty} \pi_n \quad (4)$$

We claim that $f_n(\theta) = f_{n-1}(f(\theta))$, which can be proven by the Tower property:

$$f_n(\theta) = \mathbb{E}[\mathbb{E}[\theta^{Z_n} \mid Z_{n-1}]] \quad (5)$$

$$= \mathbb{E}[\mathbb{E}[\theta^{X_1^{(n)} + \dots + X_{Z_{n-1}}^{(n)}} \mid Z_{n-1}]] \quad (6)$$

$$= \mathbb{E}[f(\theta)^{Z_{n-1}}] \quad (7)$$

$$= f_{n-1}(f(\theta)) \quad (8)$$

where the final line is true since $f_{n-1}(\alpha) = \mathbb{E}[\alpha^{Z_{n-1}}]$. This gives us a nice recursive formula, which implies that

$$\pi_n = f_n(0) = f_{n-1}(f(0)) \quad (9)$$

$$= \underbrace{(f \circ \dots \circ f)}_{n \text{ times}}(0) \quad (10)$$

$$= f(f_{n-1}(0)) \quad (11)$$

$$= f(\pi_{n-1}) \quad (12)$$

So π satisfies $\pi = f(\pi)$ as long as f is continuous.

So let's actually do some computation.

1. Assume $X \sim \text{Geom}(p)$ i.e. $\mathbb{P}(X = k) = (1-p)^k p$ for $0 < p < 1$. Then,

$$f(\theta) = \frac{p}{1 - (1-p)\theta} \quad (13)$$

Since $\theta = f(\theta)$, we can solve

$$\theta = \frac{p}{1 - (1-p)\theta} \implies \theta = 1 \text{ or } \theta = \frac{p}{1-p} \quad (14)$$

So how do we know which solution is p ? If $p \geq 1/2$, then $\pi = 1$, but if $\frac{p}{p-1} < 1$, then we can derive further

$$f_n(0) = \frac{p\left(\frac{1-p}{p}\right) - p}{(1-p)\left(\frac{1-p}{p}\right)^n - p} \rightarrow \frac{p}{1-p} \quad (15)$$

as $n \rightarrow \infty$, so $\pi = \frac{p}{1-p}$ is the correct extinction probability.

It turns out that according to some deeper theorem, if $\mathbb{E}[X] > 1$, then π is the unique solution to $x = f(x)$ on $(0, 1)$, and if $\mathbb{E}[X] \leq 1$, then $\pi = 1$.

Example 1.2

We can describe Z_n more precisely as $n \rightarrow \infty$ using a Markov chain, which can be stated in the equivalent ways.

1.

$$\mathbb{P}(Z_{n+1} = j \mid Z_0 = i_0, \dots, Z_n = i_n) = \mathbb{P}(Z_{n+1} = j \mid Z_n = i_n) \quad (16)$$

2.

$$\mathbb{E}[Z_{n+1} \mid Z_0, \dots, Z_n] = \mathbb{E}[Z_{n+1} \mid Z_n] \quad (17)$$

Observe that $M_n = Z_n/\mu^n$ is a martingale, with $\mu = \mathbb{E}[X]$, i.e. $\mathbb{E}[M_n \mid M_{n-1}] = M_{n-1}$. So,

$$\mathbb{E}[M_n \mid M_{n-1}] = \mathbb{E}[Z_n \mid Z_{n-1}] = \mu \frac{Z_{n-1}}{\mu_n} = \frac{Z_{n-1}}{\mu^{n-1}} = M_{n-1} \quad (18)$$

By a Martingale convergence theorem, setting $M_\infty(\omega) = \lim_{n \rightarrow \infty} M_n(\omega)$, we have $M_n \rightarrow M_\infty$ a.s.

Next time, we'll review conditional expectation, then martingales, then markov chains, then ergodic theory (building on convergence results), and finally with Brownian motion.

2 Jan 14

Definition 2.1 (Conditional Expectation)

Given probability space $(\Omega, \mathcal{F}_0, \mathbb{P})$, a σ -algebra $\mathcal{F} \subseteq \mathcal{F}_0$ and a random variable X with $\mathbb{E}[X] < +\infty$, we define the **conditional expectation** of X given $\mathcal{F}$ —denoted $\mathbb{E}[X \mid \mathcal{F}]$ —as *any* random variable (called a **version**) $Y : \Omega \rightarrow \mathbb{R}$ s.t.

1. Y is $\mathcal{F}$ -measurable.
2. For all $A \in \mathcal{F}$, $\int_A X d\mathbb{P} = \int_A Y d\mathbb{P}$

Proof. We should claim that it always exists and it is unique up to measure 0.

1. *Uniqueness* Show that if Y is a version, then $\mathbb{E}[|Y|] < +\infty$. Hint: Show that $\mathbb{E}[|Y|] < \mathbb{E}[|X|]$. We split $Y = Y_+ - Y_-$. Then, $\mathbb{E}[|Y|] = \mathbb{E}[Y_+] + \mathbb{E}[Y_-]$, and we can bound

$$\mathbb{E}[Y_+] = \int_{\mathbb{R}_{>0}} Y_+ d\mathbb{P} = \int_{\mathbb{R}_{>0}} X d\mathbb{P} \leq \int_{\mathbb{R}_{\geq 0}} |X| d\mathbb{P} \quad (19)$$

$$\mathbb{E}[Y_-] = \int_{\mathbb{R}_{<0}} Y_- d\mathbb{P} = \int_{\mathbb{R}_{<0}} X d\mathbb{P} \leq \int_{\mathbb{R}_{<0}} |X| d\mathbb{P} \quad (20)$$

Combining these together, we get $\mathbb{E}[|Y|] < \mathbb{E}[|X|]$. To prove uniqueness, we can see that if Y, Y' are two versions of $\mathbb{E}[X | \mathcal{F}]$, then let $A_\epsilon = \{\omega \in \Omega: Y(\omega) - Y'(\omega) > \epsilon\}$. Then,

$$\mathbb{P}(A)\epsilon \leq \int_A (Y - Y') d\mathbb{P} = \int_A (X - X) = 0 \implies Y \leq Y' \text{ a.s.} \quad (21)$$

Now swap Y and Y' to get $Y' \leq Y$.

2. *Existence.* To prove existence of Y , this is much more involved. Recall the definition of absolutely continuous measures and σ -finite measures, along with the Radon-Nikodym theorem. We will use this to construct our conditional expectation. Since we can write $X = X_+ - X_-$, WLOG let $X \geq 0$. Let $\mu = \mathbb{P}$, we are looking for Y s.t. for all $A \in \mathcal{F} \subset \mathcal{F}_0$,

$$\int_A Y d\mu = \int_A X d\mu \quad (22)$$

Now set $\nu(A) := \int_A X d\mu$, which turns out to be a σ -finite measure on $(\Omega, \mathcal{F})$. By construction, $\nu \ll \mu$. By the Radon-Nikodym theorem, $\exists f$ s.t. for all $A \in \mathcal{F}$,

$$\int_A f d\mu = \nu(A) = \int_A X d\mu \quad (23)$$

Now set $f = Y$.

So it is the best approximation in a sense. The motivation is that it is a projection of X onto the space of $\mathcal{F}$ -measurable random variables.

Example 2.1

Suppose X, Z are random variables with a joint pdf $f_{X,Z}(x, z)$ and $\mathbb{E}[|X|] < +\infty$. Then, we define the conditional pdf as

$$f_{X|Z}(x | z) := \begin{cases} \frac{f_{X,Z}(x, z)}{f_Z(z)} & \text{if } f_Z(z) \neq 0 \\ 0 & \text{if } f_Z(z) = 0 \end{cases} \quad (24)$$

Define $Y(z) := \int_{\mathbb{R}} x f_{X|Z}(x | z) dx$. We claim that $Y(Z)$ is a version of $\mathbb{E}[X | \sigma(Z)]$. As a proof, for all $A \in \sigma(Z)$, we have

$$\int_A Y(z) f_Z(z) dz = \int_A \left(\int_{\mathbb{R}} x f_{X|Z}(x | z) dx \right) f_Z(z) dz \quad (25)$$

$$= \int_A \int_{\mathbb{R}} x f_{X,Z}(x, z) dx dz \quad (26)$$

where we denote dx, dz are the Lebesgue or counting measure, depending on whether it is discrete or continuous.

Now we list all of the properties of conditional expectation.

Theorem 2.1 (Tower Rule)

Let X be a random variable on $(\Omega, \mathcal{F}, \mathbb{P})$ with $\mathbb{E}[|X|] < +\infty$. Let $\mathcal{G}, \mathcal{H}$ be sub- σ -algebras of $\mathcal{F}$. If Y is any version of $\mathbb{E}[X | \mathcal{G}]$, then $\mathbb{E}[X] = \mathbb{E}[Y]$.

Proof. $\Omega \in \mathcal{G}$ implies that

$$\mathbb{E}[Y] = \int_{\Omega} Y \, d\mathbb{P} = \int_{\Omega} X \, d\mathbb{P} = \mathbb{E}[X] \quad (27)$$

Theorem 2.2

If X is $\mathcal{G}$ -measurable, then $\mathbb{E}[X \mid \mathcal{G}] = X$ a.s.

Proof. X is a version of $\mathbb{E}[X \mid \mathcal{G}]$ because X is $\mathcal{G}$ -measurable.

Theorem 2.3 (Linearity)

We have

$$\mathbb{E}[aX_1 + bX_2 \mid \mathcal{G}] = a\mathbb{E}[X_1 \mid \mathcal{G}] + b\mathbb{E}[X_2 \mid \mathcal{G}] \quad (28)$$

Proof. We use linearity of the integral. For all $A \in \mathcal{G}$,

$$\int_A (aX_1 + bX_2) \, d\mathbb{P} = a \int_A X_1 \, d\mathbb{P} + b \int_A X_2 \, d\mathbb{P} = a \int_A \mathbb{E}[X_1 \mid \mathcal{G}] \, d\mathbb{P} + b \int_A \mathbb{E}[X_2 \mid \mathcal{G}] \, d\mathbb{P} \quad (29)$$

Theorem 2.4 (Positivity)

If $X \geq 0$, then $\mathbb{E}[X \mid \mathcal{G}] \geq 0$ a.s.

Proof. Let Y be a version of $\mathbb{E}[X \mid \mathcal{G}]$. If $\mathbb{P}(Y < 0) > 0$, then there exists a $\epsilon > 0$ s.t. $\mathbb{P}(Y < -\epsilon) > 0$. Set $A = \{\omega \in \Omega \mid Y(\omega) < -\epsilon\}$. Then,

$$-\epsilon\mathbb{P}(A) > \int_A Y \, d\mathbb{P} = \int_A X \, d\mathbb{P} \geq 0 \quad (30)$$

a contradiction.

Theorem 2.5 (Conditional Monotone Convergence Theorem)

If $0 \leq X_n$ is a monotonically increasing sequence of nonnegative functions and $X_n \rightarrow X$ a.s., then

$$\mathbb{E}[X_n \mid \mathcal{G}] \nearrow \mathbb{E}[X \mid \mathcal{G}] \text{ a.s.} \quad (31)$$

Proof. Let Y_n be a version of $\mathbb{E}[X_n \mid \mathcal{G}]$. Set $Y(\omega) := \limsup_n Y_n(\omega)$, which is $\mathcal{G}$ -measurable and $Y_n \nearrow Y$. It remains to show that Y is a version of $\mathbb{E}[X \mid \mathcal{G}]$. For all $A \in \mathcal{G}$, by MCT and the fact that $Y_n \geq 0$, we have

$$\int_A Y_n \, d\mathbb{P} \rightarrow \int_A Y \, d\mathbb{P} \quad (32)$$

But since $X_n \nearrow X$, $\int_A Y_n \, d\mathbb{P} = \int_A X_n \, d\mathbb{P} \rightarrow \int_A X \, d\mathbb{P}$, so $\int_A Y \, d\mathbb{P} = \int_A X \, d\mathbb{P}$, which implies that Y is a version of $\mathbb{E}[X \mid \mathcal{G}]$.

Theorem 2.6 (Conditional Fatou's Lemma)

If $X_n \geq 0$, then

$$\mathbb{E}[\liminf_n X_n \mid \mathcal{G}] \leq \liminf_n \mathbb{E}[X_n \mid \mathcal{G}] \text{ a.s.} \quad (33)$$

Theorem 2.7 (Conditional Dominated Convergence Theorem)

If $|X_n| \leq V$ for all $n \in \mathbb{N}$ and $\mathbb{E}[|V|] < +\infty$, and $X_n \rightarrow X$ a.s., then $\mathbb{E}[X_n \mid \mathcal{G}] \rightarrow \mathbb{E}[X \mid \mathcal{G}]$ a.s.

Theorem 2.8 (Conditional Jensen's Inequality)

If $f : \mathbb{R} \rightarrow \mathbb{R}$ is convex, and $\mathbb{E}[|f(X)|] < +\infty$, then

$$f(\mathbb{E}[X \mid \mathcal{G}]) \leq \mathbb{E}[f(X) \mid \mathcal{G}] \quad (34)$$

As a corollary, we have $\|\mathbb{E}[X \mid \mathcal{G}]\|_p \leq \|X\|_p$ for $p \geq 1$, so conditional expectation is a contraction.

Proof. use the fact that $f(x) = \sup_n (a_n x + b_n)$ for some sequence $\{(a_n, b_n)\}_n$, and then invoke linearity.

Theorem 2.9 (Tower Rule)

If $\mathcal{H}$ is a sub- σ -algebra of $\mathcal{G}$, then

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}] \mid \mathcal{H}] = \mathbb{E}[X \mid \mathcal{H}] \quad (35)$$

So, the smaller σ -algebra always wins.

Theorem 2.10 (Taking Out What's Known)

If Z is $\mathcal{G}$ -measurable and bounded, then

$$\mathbb{E}[ZX \mid \mathcal{G}] = Z\mathbb{E}[X \mid \mathcal{G}] \quad (36)$$

Proof. Start out with indicators, then simple, then bounded as pointwise limits of simple functions.

Theorem 2.11 (Role of Independence)

If $\mathcal{H}$ is independent of $\sigma(\mathcal{G}, X)$, then

$$\mathbb{E}[X \mid \sigma(\mathcal{G}, \mathcal{H})] = \mathbb{E}[X \mid \mathcal{G}] \text{ a.s.} \quad (37)$$

In particular, $\mathbb{E}[X \mid \mathcal{H}] = \mathbb{E}[X]$ a.s.

Proof. Use π -systems and such, in Durrett's book.